

41. Let the point, on the line passing through the points $P(1, -2, 3)$ and $Q(5, -4, 7)$, farther from the origin and at a distance of 9 units from the point P , be (α, β, γ) . Then $\alpha^2 + \beta^2 + \gamma^2$ is equal to :

माना बिन्दुओं $P(1, -2, 3)$ तथा $Q(5, -4, 7)$ से होकर जाने वाली रेखा पर, मूल बिन्दु से ज्यादा दूर तथा बिन्दु P से 9 इकाई दूरी पर, स्थित बिन्दु (α, β, γ) है, तो $\alpha^2 + \beta^2 + \gamma^2$ बराबर है :

- (1) 155 (2) 150 (3) 165 (4) 160

Ans. A

Sol. PQ line

$$\frac{x-1}{4} = \frac{y+2}{-2} = \frac{z-3}{4}$$

$$\text{pt}(4t+1, -2t-2, 4t+3)$$

$$\text{distance}^2 = 16t^2 + 4t^2 + 16t^2 = 81$$

$$t = \pm \frac{3}{2}$$

$$\text{pt}(7, -5, 9)$$

$$\alpha^2 + \beta^2 + \gamma^2 = 155$$

42. Let the line L intersect the lines $x-2 = -y = z-1$, $2(x+1) = 2(y-1) = z+1$ and be parallel to the line

$$\frac{x-2}{3} = \frac{y-1}{1} = \frac{z-2}{2}. \text{ Then which of the following points lies on } L?$$

माना रेखा L , रेखाओं $x-2 = -y = z-1$, $2(x+1) = 2(y-1) = z+1$ को काटती है, तथा रेखा $\frac{x-2}{3} = \frac{y-1}{1} = \frac{z-2}{2}$ के

समान्तर है। तो निम्न में से कौनसा बिन्दु L पर है?

- (1) $\left(-\frac{1}{3}, -1, -1\right)$ (2) $\left(-\frac{1}{3}, -1, 1\right)$ (3) $\left(-\frac{1}{3}, 1, -1\right)$ (4) $\left(-\frac{1}{3}, 1, 1\right)$

Ans. C

Sol. $\frac{x-2}{1} = \frac{y}{-1} = \frac{z-1}{1}, \frac{x+1}{1/2} = \frac{y-1}{1/2} = \frac{z+1}{1}$

$$(k+2, -k, k+1), \left(\frac{\alpha}{2}-1, \frac{\alpha}{2}+1, \alpha-1\right)$$

$$\frac{\frac{\alpha}{2}-1-k-2}{3} = \frac{\frac{\alpha}{2}+1+k}{1} = \frac{\alpha-1-k-1}{2}$$

$$k = -\frac{4}{3}$$

∴ L passes through point $\left(\frac{2}{3}, \frac{4}{3}, -\frac{1}{3}\right)$.

$$L: \frac{x - \frac{2}{3}}{\frac{2}{3}} = \frac{y - \frac{4}{3}}{1} = \frac{z + \frac{1}{3}}{2}$$

$\left(-\frac{1}{3}, 1, -1\right)$ lies on L

43. Let $\vec{a} = \hat{i} + 2\hat{j} + \hat{k}$, $\vec{b} = 3(\hat{i} - \hat{j} + \hat{k})$. Let $\vec{c}$ be the vector such that $\vec{a} \times \vec{c} = \vec{b}$ and $\vec{a} \cdot \vec{c} = 3$. Then $\vec{a} \cdot ((\vec{c} \times \vec{b}) - \vec{b} - \vec{c})$ is equal to :

माना $\vec{a} = \hat{i} + 2\hat{j} + \hat{k}$, $\vec{b} = 3(\hat{i} - \hat{j} + \hat{k})$ है। माना एक सदिश $\vec{c}$ के लिए $\vec{a} \times \vec{c} = \vec{b}$ तथा $\vec{a} \cdot \vec{c} = 3$ हैं। तो $\vec{a} \cdot ((\vec{c} \times \vec{b}) - \vec{b} - \vec{c})$ बराबर है :

- (1) 36 (2) 24 (3) 32 (4) 20

Ans. B

Sol. $(\vec{a} \times \vec{c}) \cdot \vec{b} = \vec{b} \cdot \vec{b}$

$$[\vec{a} \vec{c} \vec{b}] = 9(1+1+1) = 27$$

$$\vec{a} \cdot \vec{b} = 3(1-2+1) = 0$$

$$\vec{a} \cdot \vec{c} = 3$$

$$\vec{a} \cdot ((\vec{c} \times \vec{b}) - \vec{b} - \vec{c}) = [\vec{a} \vec{c} \vec{b}] - \vec{a} \cdot \vec{b} - \vec{a} \cdot \vec{c}$$

$$= 27 - 0 - 3 = 24$$

44. Let the position vectors of the vertices A, B and C of a triangle be $2\hat{i} + 2\hat{j} + \hat{k}$, $\hat{i} + 2\hat{j} + 2\hat{k}$ and $2\hat{i} + \hat{j} + 2\hat{k}$ respectively. Let ℓ_1, ℓ_2 and ℓ_3 be the lengths of perpendiculars drawn from the ortho center of the triangle on the sides AB, BC and CA respectively, then $\ell_1^2 + \ell_2^2 + \ell_3^2$ equal :

माना एक त्रिभुज के शीर्ष A, B तथा C के स्थिति सदिश क्रमशः $2\hat{i} + 2\hat{j} + \hat{k}$, $\hat{i} + 2\hat{j} + 2\hat{k}$ तथा $2\hat{i} + \hat{j} + 2\hat{k}$ हैं। माना इस त्रिभुज के लम्ब केन्द्र से भुजाओं AB, BC तथा CA पर डाले गए लम्बों की लम्बाइयाँ क्रमशः ℓ_1, ℓ_2 तथा ℓ_3 हैं, तो $\ell_1^2 + \ell_2^2 + \ell_3^2$ बराबर है :

- (1) $\frac{1}{5}$ (2) $\frac{1}{2}$ (3) $\frac{1}{4}$ (4) $\frac{1}{3}$

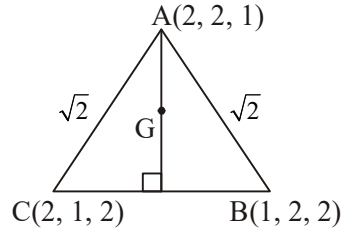
Ans. B

Sol. Equilateral Δ

$$\sqrt{2 - \frac{1}{2}} = \sqrt{\frac{3}{2}}$$

$$\frac{1}{3} \times \sqrt{\frac{3}{2}} = \frac{1}{\sqrt{6}}$$

$$\ell_1^2 + \ell_2^2 + \ell_3^2 = 3\ell_1^2 = 3 \times \frac{1}{6} = \frac{1}{2}$$



45. The position vectors of the vertices A, B and C of a triangle are $2\hat{i} - 3\hat{j} + 3\hat{k}$, $2\hat{i} + 2\hat{j} + 3\hat{k}$ and $-\hat{i} + \hat{j} + 3\hat{k}$ respectively. Let l denotes the length of the angle bisector AD of $\angle BAC$ where D is on the line segment BC, then $2l^2$ equals :

एक त्रिभुज के शीर्षों A, B तथा C के स्थिति सदिश क्रमशः $2\hat{i} - 3\hat{j} + 3\hat{k}$, $2\hat{i} + 2\hat{j} + 3\hat{k}$ तथा $-\hat{i} + \hat{j} + 3\hat{k}$ हैं। माना कोण $\angle BAC$ के समद्विभाजक AD की लम्बाई l है, जहाँ बिंदु D, रेखाखण्ड BC पर है, तो $2l^2$ बराबर है :

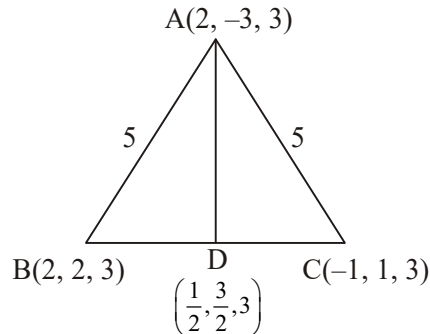
- (1) 49 (2) 42 (3) 50 (4) 45

Ans. D

Sol. $\ell^2 = \left(2 - \frac{1}{2}\right)^2 + \left(3 + \frac{3}{2}\right)^2 + 0$

$$\ell^2 = \frac{9}{4} + \frac{81}{4}$$

$$\ell^2 = \frac{45}{2} \quad 2\ell^2 = 45$$



46. Let O be the origin and the position vectors of A and B be $2\hat{i} + 2\hat{j} + \hat{k}$ and $2\hat{i} + 4\hat{j} + 4\hat{k}$ respectively. If the internal bisector of $\angle AOB$ meets the line AB at C, then the length of OC is :

माना O मूलबिन्दु है तथा A और B के स्थिति सदिश क्रमशः $2\hat{i} + 2\hat{j} + \hat{k}$ और $2\hat{i} + 4\hat{j} + 4\hat{k}$ हैं। यदि $\angle AOB$ की अंतः समद्विभाजक रेखा AB को C पर मिलती है, तो OC की लंबाई है :

- (1) $\frac{2}{3}\sqrt{34}$ (2) $\frac{3}{2}\sqrt{31}$ (3) $\frac{2}{3}\sqrt{31}$ (4) $\frac{3}{2}\sqrt{34}$

Ans. A

Sol. $|\vec{OA}| = \sqrt{4+4+1} = 3$

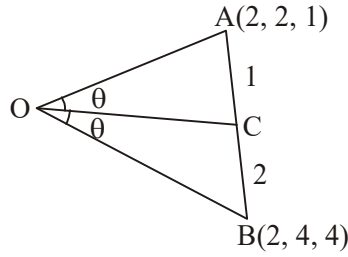
$$|\overrightarrow{OB}| = \sqrt{4+16+16} = 6$$

$$AC : CB = 3 : 6 = 1 : 2$$

$$c\left(\frac{4+2}{3}, \frac{4+4}{3}, \frac{2+4}{3}\right)$$

$$c\left(2, \frac{8}{3}, 2\right)$$

$$|\overrightarrow{OC}| = \sqrt{4 + \frac{64}{9} + 4} = \frac{\sqrt{136}}{3} = \frac{2}{3}\sqrt{34}$$



47. Let a unit vector $\vec{u} = x\hat{i} + y\hat{j} + z\hat{k}$ make angles $\frac{\pi}{2}$, $\frac{\pi}{3}$ and $\frac{2\pi}{3}$ with the vectors $\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{k}$, $\frac{1}{\sqrt{2}}\hat{j} + \frac{1}{\sqrt{2}}\hat{k}$ and $\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$ respectively. If $\vec{v} = \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j} + \frac{1}{\sqrt{2}}\hat{k}$ then $|\vec{u} - \vec{v}|^2$ is equal to :

माना एक मात्रक सदिश $\vec{u} = x\hat{i} + y\hat{j} + z\hat{k}$, सदिशों $\frac{1}{\sqrt{2}}\hat{j} + \frac{1}{\sqrt{2}}\hat{k}$, $\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$ तथा $\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{k}$ से क्रमशः

$\frac{\pi}{2}$, $\frac{\pi}{3}$ तथा $\frac{2\pi}{3}$ के कोण बनाता है। यदि $\vec{v} = \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j} + \frac{1}{\sqrt{2}}\hat{k}$ है, तो $|\vec{u} - \vec{v}|^2$ बराबर है :

- (1) $\frac{5}{2}$ (2) 7 (3) $\frac{11}{2}$ (4) 9

Ans. A

Sol. Unit vector $\vec{u} = x\hat{i} + y\hat{j} + z\hat{k}$

$$\vec{p}_1 = \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{k}, \vec{p}_2 = \frac{1}{\sqrt{2}}\hat{j} + \frac{1}{\sqrt{2}}\hat{k}$$

$$\vec{p}_3 = \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$$

Now angle between $\vec{u}$ and $\vec{p}_1 = \frac{\pi}{2}$

$$\vec{u} \cdot \vec{p}_1 = 0 \Rightarrow \frac{x}{\sqrt{2}} + \frac{z}{\sqrt{2}} = 0$$

$$\Rightarrow x + z = 0$$

Now angle between $\vec{u}$ and $\vec{p}_2 = \frac{\pi}{3}$

$$\hat{u} \cdot \vec{p}_2 = |\hat{u}| \cdot |\vec{p}_2| \cos \frac{\pi}{3}$$

$$\Rightarrow \frac{y}{\sqrt{2}} + \frac{z}{\sqrt{2}} = \frac{1}{2} \Rightarrow y + z = \frac{1}{\sqrt{2}}$$

Now angle between $\hat{u}$ and $\vec{p}_3 = \frac{2\pi}{3}$

$$\hat{u} \cdot \vec{p}_3 = |\hat{u}| \cdot |\vec{p}_3| \cos \frac{2\pi}{3}$$

$$\Rightarrow \frac{x}{\sqrt{2}} + \frac{4}{\sqrt{2}} = \frac{-1}{2}$$

$$\Rightarrow x + y = \frac{-1}{\sqrt{2}}$$

from equation (i), (ii) and (iii) we get

$$x = \frac{-1}{\sqrt{2}} \quad y = 0 \quad z = \frac{1}{\sqrt{2}}$$

$$\text{Thus } \hat{u} - \vec{v} = \frac{-1}{\sqrt{2}} \hat{i} + \frac{1}{\sqrt{2}} \hat{k} - \frac{1}{\sqrt{2}} \hat{i} - \frac{1}{\sqrt{2}} \hat{j} - \frac{1}{\sqrt{2}} \hat{k}$$

$$\hat{u} - \vec{v} = \frac{-2}{\sqrt{2}} \hat{i} - \frac{1}{\sqrt{2}} \hat{j}$$

$$\therefore |\hat{u} - \vec{v}|^2 = \left(\sqrt{\frac{4}{2} + \frac{1}{2}} \right)^2 = \frac{5}{2}$$

48. Let $\vec{a} = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}$ and $\vec{b} = b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}$ be two vectors such that $|\vec{a}| = 1$, $\vec{a} \cdot \vec{b} = 2$ and $|\vec{b}| = 4$.

If $\vec{c} = 2(\vec{a} \times \vec{b}) - 3\vec{b}$, then the angle between $\vec{b}$ and $\vec{c}$ is equal to :

माना दो सदिशों $\vec{a} = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}$ तथा $\vec{b} = b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}$ के लिए $|\vec{a}| = 1$, $\vec{a} \cdot \vec{b} = 2$ तथा $|\vec{b}| = 4$ हैं। यदि

$\vec{c} = 2(\vec{a} \times \vec{b}) - 3\vec{b}$ है, तो $\vec{b}$ तथा $\vec{c}$ के बीच कोण है :

$$(1) \cos^{-1}\left(-\frac{1}{\sqrt{3}}\right) \quad (2) \cos^{-1}\left(\frac{2}{3}\right) \quad (3) \cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) \quad (4) \cos^{-1}\left(\frac{2}{\sqrt{3}}\right)$$

Ans. C

Sol. Given $|\vec{a}| = 1$, $|\vec{b}| = 4$, $\vec{a} \cdot \vec{b} = 2$

$$\vec{c} = 2(\vec{a} \times \vec{b}) - 3\vec{b}$$

Dot product with $\vec{a}$ on both sides

$$\vec{c} \cdot \vec{a} = -6 \quad \dots(1)$$

Dot product with $\vec{b}$ on both sides

$$\vec{b} \cdot \vec{c} = -48 \quad \dots(2)$$

$$\vec{c} \cdot \vec{c} = 4|\vec{a} \times \vec{b}| + 9|\vec{b}|^2$$

$$|\vec{c}|^2 = 4\left[|\vec{a}|^2|\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2\right] + 9|\vec{b}|^2$$

$$|\vec{c}|^2 = 4\left[(1)(4)^2 - (4)^2\right] + 9(16)$$

$$|\vec{c}|^2 = 4[12] + 144$$

$$|\vec{c}|^2 = 192$$

$$\therefore \cos \theta = \frac{\vec{b} \cdot \vec{c}}{|\vec{b}||\vec{c}|}$$

$$\therefore \cos \theta = \frac{-48}{\sqrt{192} \cdot 4}$$

$$\therefore \cos \theta = \frac{-48}{8\sqrt{3} \cdot 4}$$

$$\therefore \cos \theta = \frac{-3}{2\sqrt{3}}$$

$$\therefore \cos \theta = \frac{-\sqrt{3}}{2} \Rightarrow \theta = \cos^{-1}\left(\frac{-\sqrt{3}}{2}\right)$$

49. Let $\vec{a}$ and $\vec{b}$ be two vectors such that $|\vec{b}| = 1$ and $|\vec{b} \times \vec{a}| = 2$. Then $\left|(\vec{b} \times \vec{a}) - \vec{b}\right|^2$ is equal to :

माना दो सदिश $\vec{a}$ तथा $\vec{b}$ इस प्रकार हैं कि $|\vec{b}| = 1$ तथा $|\vec{b} \times \vec{a}| = 2$ हैं। तो $\left|(\vec{b} \times \vec{a}) - \vec{b}\right|^2$ बराबर है :

- (1) 3 (2) 1 (3) 5 (4) 4

Ans. C

Sol.
$$\begin{aligned} &= |\vec{b} \times \vec{a}|^2 + |\vec{b}|^2 - 2(\vec{b} \times \vec{a}) \cdot \vec{b} \\ &= 4 + 1 - 0 = 5 \end{aligned}$$

50. Let $\vec{a} = \hat{i} + \alpha\hat{j} + \beta\hat{k}, \alpha, \beta \in \mathbb{R}$. Let a vector $\vec{b}$ be such that the angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{4}$ and $|\vec{b}|^2 = 6$. If $\vec{a} \cdot \vec{b} = 3\sqrt{2}$, then the value of $(\alpha^2 + \beta^2)|\vec{a} \times \vec{b}|^2$ is :

माना $\vec{a} = \hat{i} + \alpha\hat{j} + \beta\hat{k}, \alpha, \beta \in \mathbb{R}$ है। माना एक सदिश $\vec{b}$ इस प्रकार है कि $\vec{a}$ तथा $\vec{b}$ के बीच कोण $\frac{\pi}{4}$ है तथा $|\vec{b}|^2 = 6$ है। यदि

$\vec{a} \cdot \vec{b} = 3\sqrt{2}$ है, तो $(\alpha^2 + \beta^2)|\vec{a} \times \vec{b}|^2$ बराबर है :

- (1) 75 (2) 95 (3) 85 (4) 90

Ans. D

Sol. $\vec{a} \cdot \vec{b} = 3\sqrt{2}$

$$|\vec{a}||\vec{b}|\cos\frac{\pi}{4} = 3\sqrt{2}$$

$$|\vec{a}|\sqrt{6} \cdot \frac{1}{\sqrt{2}} = 3\sqrt{2}$$

$$|\vec{a}| = \sqrt{6} = \sqrt{1 + \alpha^2 + \beta^2}$$

$$\alpha^2 + \beta^2 = 5$$

$$|\vec{a} \times \vec{b}|^2 = |\vec{a}|^2 |\vec{b}|^2 \sin^2 \frac{\pi}{4}$$

$$= 6 \times 6 \times \frac{1}{2} = 18$$

$$\text{So, } (\alpha^2 + \beta^2)|\vec{a} \times \vec{b}|^2 = 5 \times 18 = 90$$

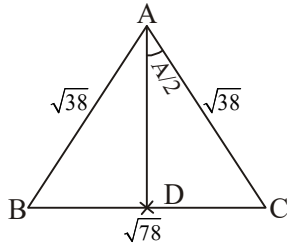
51. Consider a ΔABC where $A(1, 3, 2)$, $B(-2, 8, 0)$ and $C(3, 6, 7)$. If the angle bisector of $\angle BAC$ meets the line BC at D , then the length of the projection of the vector $\overrightarrow{AD}$ on the vector $\overrightarrow{AC}$ is :

एक ΔABC , जहाँ $A(1, 3, 2)$, $B(-2, 8, 0)$ तथा $C(3, 6, 7)$ हैं, का विचार कीजिए। यदि कोण BAC की समद्विभाजक रेखा, BC को D पर मिलती है, तो सदिश $\overrightarrow{AD}$ के सदिश $\overrightarrow{AC}$ पर प्रक्षेप की लम्बाई है :

- (1) $\frac{37}{2\sqrt{38}}$ (2) $\sqrt{19}$ (3) $\frac{\sqrt{38}}{2}$ (4) $\frac{39}{2\sqrt{38}}$

Ans. A

Sol.



$$\begin{aligned}
 \text{Length of projection of } \overrightarrow{AD} \text{ a long } \overrightarrow{AC} &= AD \cos \frac{A}{2} \\
 &= \left(\frac{2bc}{b+c} \right) \cos^2 \frac{A}{2} \\
 &= \left(\frac{2bc}{b+c} \right) \left(\frac{1+\cos A}{2} \right) \\
 &= \frac{37}{2\sqrt{38}}
 \end{aligned}$$

Method -II

$$\begin{aligned}
 D &\left(\frac{1}{2}, 7, \frac{7}{2} \right) \\
 \Rightarrow \overrightarrow{AD} &= \frac{\hat{i} - 8\hat{j} - 3\hat{k}}{2} \\
 \text{Length} &= \left| \frac{\overrightarrow{AD} \cdot \overrightarrow{AC}}{|\overrightarrow{AC}|} \right| = \frac{37}{2\sqrt{38}}
 \end{aligned}$$

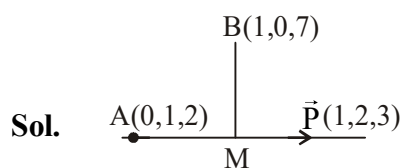
52. Let the image of the point $(1,0,7)$ in the line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ be the point (α, β, γ) Then which one of the following points lies on the line passing through (α, β, γ) and making angles $\frac{2\pi}{3}$ and $\frac{3\pi}{4}$ with y-axis and z-axis respectively and an acute angle with x-axis ?

माना रेखा $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ में बिन्दु $(1,0,7)$ का प्रतिबिम्ब (α, β, γ) है। तो निम्न में से कौनसा बिंदु, बिंदु (α, β, γ) से होकर

जाने वाली, y और z-अक्षों से क्रमशः $\frac{2\pi}{3}$ तथा $\frac{3\pi}{4}$ का कोण बनाने वाली तथा x-अक्ष से न्यून कोण बनाने वाली रेखा पर है?

- (1) $(1, 2, 1 - \sqrt{2})$ (2) $(3, -4, 3 + 2\sqrt{2})$ (3) $(1, -2, 1 + \sqrt{2})$ (4) $(3, 4, 3 - 2\sqrt{2})$

Ans. D



$$M = (\lambda, 1 + 2\lambda, 2 + 3\lambda)$$

$$\overrightarrow{BM} = (\lambda - 1, 2\lambda + 1, 3\lambda - 5)$$

$$\vec{P} = (1, 2, 3)$$

$$\overrightarrow{BM} \cdot \vec{P} = 0$$

$$1(\lambda - 1) + 2(2\lambda + 1) + 3(3\lambda - 5) = 0$$

$$\lambda = 1$$

$$M = (1, 3, 5)$$

$$\text{Image } (1, 6, 3)$$

$$m = \cos 2\pi / 3 = -1/2$$

$$n = \cos 3\pi / 4 = -\frac{1}{\sqrt{2}}$$

$$m^2 + n^2 + \ell^2 = 1$$

$$\ell^2 = \frac{1}{4}$$

$$\ell = \pm \frac{1}{2}$$

So, required line is

$$\frac{x-1}{\pm \frac{1}{2}} = \frac{y-6}{-\frac{1}{2}} = \frac{z-3}{-\frac{1}{\sqrt{2}}}$$

$$\frac{x-1}{\pm 1} = \frac{y-6}{1} = \frac{z-3}{\sqrt{2}}$$

53. Let PQR be a triangle with R (-1, 4, 2). Suppose M (2, 1, 2) is the mid point of PQ. The distance of the centroid of ΔPQR from the point of intersection of the lines $\frac{x-2}{0} = \frac{y}{2} = \frac{z+3}{-1}$ and $\frac{x-1}{1} = \frac{y+3}{-3} = \frac{z+1}{1}$ is:

माना एक त्रिभुज PQR में R (-1, 4, 2) है। माना PQ का मध्य बिन्दु M (2, 1, 2) है। तो रेखाओं $\frac{x-2}{0} = \frac{y}{2} = \frac{z+3}{-1}$ तथा $\frac{x-1}{1} = \frac{y+3}{-3} = \frac{z+1}{1}$ के प्रतिच्छेद बिन्दु से त्रिभुज PQR के केंद्र के दूरी ज्ञात करें।

$\frac{x-1}{1} = \frac{y+3}{-3} = \frac{z+1}{1}$ के प्रतिच्छेदन बिन्दु से ΔPQR के केन्द्रक की दूरी है :

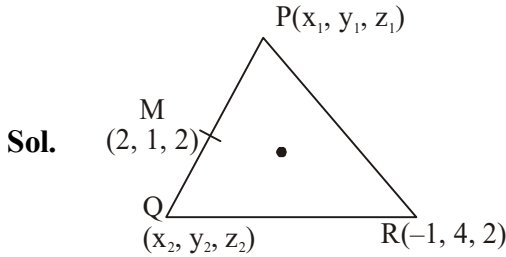
(1) $\sqrt{99}$

(2) $\sqrt{69}$

(3) 69

(4) 9

Ans. B



Given two lines are

$$\frac{x-2}{0} = \frac{y}{2} = \frac{z+3}{-1} = \lambda \text{ (let)}$$

$$\frac{x-1}{1} = \frac{y+3}{-3} = \frac{z+1}{1} = \mu \text{ (let)}$$

for point of intersection

$$x = 2, y = 2\lambda, z = -\lambda - 3 \text{ (General point of line 1)}$$

$$x = \mu + 1, y = -3\mu - 3, z = \mu - 1 \text{ (General point of line 2)}$$

$$2 = \mu + 1 \Rightarrow \mu = 1$$

$$2\lambda = -3\mu - 3 \Rightarrow \lambda = -3$$

Point of intersection of two lines is $(2, -6, 0)$

$\therefore$ M is the mid point of PQ,

$$\therefore x_1 + x_2 = 4, y_1 + y_2 = 2, z_1 + z_2 = 4$$

Centroid of ΔPQR

$$= \left(\frac{x_1 + x_2 - 1}{3}, \frac{y_1 + y_2 + 4}{3}, \frac{z_1 + z_2 + 2}{3} \right)$$

$$= (1, 2, 2)$$

$\therefore$ Distance between $(1, 2, 2)$ and $(2, -6, 0)$

$$= \sqrt{1^2 + 8^2 + 2^2} = \sqrt{69}$$

54. Let P and Q be the points on the line $\frac{x+3}{8} = \frac{y-4}{2} = \frac{z+1}{2}$ which are at a distance of 6 units from the point R(1, 2, 3). If the centroid of the triangle PQR is (α, β, γ) , then $\alpha^2 + \beta^2 + \gamma^2$ is :

माना रेखा $\frac{x+3}{8} = \frac{y-4}{2} = \frac{z+1}{2}$ पर बिंदु R(1, 2, 3) से 6 इकाई दूरी पर दो बिंदु P तथा Q हैं। यदि त्रिभुज PQR का केन्द्रक $(\alpha,$

β, γ है, तो $\alpha^2 + \beta^2 + \gamma^2$ बराबर है :

(1) 36

(2) 18

(3) 24

(4) 26

Ans. B

Sol. Line $\frac{x+3}{8} = \frac{y-4}{2} = \frac{z+1}{2}$

$P(8\lambda - 3, 2\lambda + 4, 2\lambda - 1)$

$PR = 6$

$\Rightarrow \lambda = 0, 1$

$\Rightarrow P(-3, 4, -1) \quad Q(5, 6, 1)$

Centroid of $\Delta PQR = (1, 4, 1)$

$\alpha = 1, \beta = 4, \gamma = 1$

$\Rightarrow \alpha^2 + \beta^2 + \gamma^2 = 18$

55. Let $\vec{a} = \hat{i} + 2\hat{j} + \hat{k}$, $\vec{b} = 3\hat{i} - 3\hat{j} + 3\hat{k}$, $\vec{c} = 2\hat{i} - \hat{j} + 2\hat{k}$ and $\vec{d}$ be a vector such that $\vec{b} \times \vec{d} = \vec{c} \times \vec{d}$ and $\vec{a} \cdot \vec{d} = 4$.

Then $\left| (\vec{a} \times \vec{d}) \right|^2$ is equal to _____.

माना $\vec{a} = \hat{i} + 2\hat{j} + \hat{k}$, $\vec{b} = 3\hat{i} - 3\hat{j} + 3\hat{k}$, $\vec{c} = 2\hat{i} - \hat{j} + 2\hat{k}$ हैं और एक सदिश $\vec{d}$ के लिए $\vec{b} \times \vec{d} = \vec{c} \times \vec{d}$ और $\vec{a} \cdot \vec{d} = 4$ हैं। तो

$\left| (\vec{a} \times \vec{d}) \right|^2$ बराबर है _____.

Ans. 128

Sol. $(\vec{b} - \vec{c}) \times \vec{d} = 0$

$\vec{b} - \vec{c} = \lambda \vec{d}$

Now taking dot with $\vec{a}$

$\vec{a} \cdot \vec{b} - \vec{a} \cdot \vec{c} = \lambda \vec{a} \cdot \vec{d}$

$\Rightarrow 0 - 2 = \lambda 4 \Rightarrow \lambda = -\frac{1}{2}$

$\vec{b} = \vec{c} + \lambda \vec{d}$

$\vec{d} = 2(\vec{c} - \vec{b}) = 2(-\hat{i} + 2\hat{j} + \hat{k}),$

$\Rightarrow |\vec{d}|^2 = 4 \times 6$

$\Rightarrow |\vec{a}|^2 = 6$

$|\vec{a} \times \vec{d}|^2 = |\vec{a}|^2 |\vec{d}|^2 - (\vec{a} \cdot \vec{d})^2$

$= 4 \times 6 \times 6 - 16 = 16(9 - 1) = 16 \times 8 = 128$

56. Let the three sides of a triangle ABC be given by the vectors $2\hat{i} - \hat{j} + \hat{k}$, $\hat{i} - 3\hat{j} - 5\hat{k}$ and $3\hat{i} - 4\hat{j} - 4\hat{k}$. Let G be the centroid of the triangle ABC. Then $6(|\overrightarrow{AG}|^2 + |\overrightarrow{BG}|^2 + |\overrightarrow{CG}|^2)$ is equal to _____.

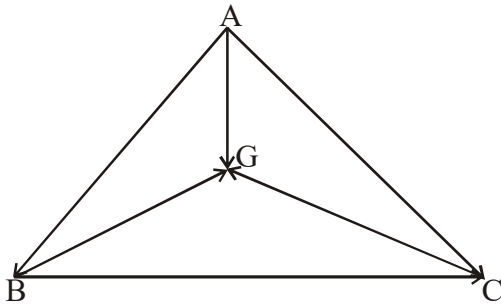
माना एक त्रिभुज ABC की तीन भुजाएँ सदिशों $2\hat{i} - \hat{j} + \hat{k}$, $\hat{i} - 3\hat{j} - 5\hat{k}$ और $3\hat{i} - 4\hat{j} - 4\hat{k}$ द्वारा दी गई हैं। माना त्रिभुज ABC का केन्द्रक G है। तो $6(|\overrightarrow{AG}|^2 + |\overrightarrow{BG}|^2 + |\overrightarrow{CG}|^2)$ बराबर है _____.

Ans. 164

Sol. $\overrightarrow{AB} = (2, -1, 1)$

$$\overrightarrow{BC} = (1, -3, -5)$$

$$\overrightarrow{AC} = (3, -4, -4)$$



$$\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$$

Let $A(0,0,0)$

then $B(2, -1, 1)$, $C(3, -4, -4)$

$$G = \left(\frac{5}{3}, -\frac{5}{3}, -1 \right)$$

$$|\overrightarrow{AG}|^2 = \frac{59}{9}$$

$$|\overrightarrow{BG}|^2 = \frac{41}{9}$$

$$|\overrightarrow{CG}|^2 = \frac{146}{9}$$

$$6\{|\overrightarrow{AG}|^2 + |\overrightarrow{BG}|^2 + |\overrightarrow{CG}|^2\}$$

$$= 164$$

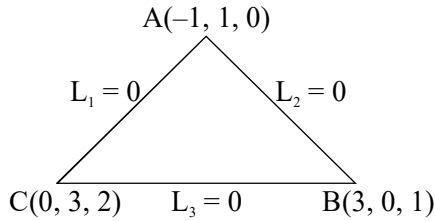
57. Let the area of the triangle formed by the lines $x + 2 = y - 1 = z$, $\frac{x-3}{5} = \frac{y}{-1} = \frac{z-1}{1}$ and $\frac{x}{-3} = \frac{y-3}{3} = \frac{z-2}{1}$ be A. Then A^2 is equal to _____.

माना रेखाओं $x + 2 = y - 1 = z$, $\frac{x-3}{5} = \frac{y}{-1} = \frac{z-1}{1}$ और $\frac{x}{-3} = \frac{y-3}{3} = \frac{z-2}{1}$ द्वारा बने त्रिभुज का क्षेत्रफल A है। तो A^2

बराबर है _____.

Ans. 56

Sol.



$$L_1 : \frac{x+2}{1} = \frac{y-1}{1} = \frac{z-0}{1} : L_2 : \frac{x-3}{5} = \frac{y}{-1} = \frac{z-1}{1} : L_3 : \frac{x-0}{-3} = \frac{y-3}{3} = \frac{z-2}{1}$$

$$A = \frac{1}{2} |\vec{AB} \times \vec{AC}| = \left| \frac{1}{2} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 5 & -1 & 1 \\ 2 & 2 & 2 \end{vmatrix} \right|$$

$$A = \frac{1}{2} |(-4\hat{i} - 8\hat{j} + 12\hat{k})|$$

$$A = \sqrt{4+16+36}$$

58. Let $\vec{c}$ be the projection vector $\vec{b} = \lambda\hat{i} + 4\hat{k}$, $\lambda > 0$, on the vector $\vec{a} = \hat{i} + 2\hat{j} + 2\hat{k}$. If $|\vec{a} + \vec{c}| = 7$, then the area of the parallelogram formed by the vectors $\vec{b}$ and $\vec{c}$ is _____.

माना कि $\vec{c}$, सदिश $\vec{b} = \lambda\hat{i} + 4\hat{k}$, $\lambda > 0$, का सदिश $\vec{a} = \hat{i} + 2\hat{j} + 2\hat{k}$ पर प्रक्षेप है। यदि $|\vec{a} + \vec{c}| = 7$ है, तो सदिशों $\vec{b}$ और $\vec{c}$ द्वारा बनाए गए समान्तर चतुर्भुज का क्षेत्रफल है _____.

Ans. 16

$$\vec{c} = \left(\frac{\vec{b} \cdot \vec{a}}{|\vec{a}|^2} \right) \vec{a}$$

$$= \left(\frac{\lambda + 8}{9} \right) (\hat{i} + 2\hat{j} + 2\hat{k})$$

$$|\vec{a} + \vec{c}| = 7 \Rightarrow \lambda = 4$$

Area of parallelogram

$$= |\vec{b} \times \vec{c}| = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & 8 & 8 \\ 3 & 3 & 3 \\ 4 & 0 & 4 \end{vmatrix}$$

$$= 16$$

59. Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = 2\hat{i} + 2\hat{j} + \hat{k}$ and $\vec{d} = \vec{a} \times \vec{b}$. If $\vec{c}$ is a vector such that $\vec{a} \cdot \vec{c} = |\vec{c}|$, $|\vec{c} - 2\vec{a}|^2 = 8$ and the angle between $\vec{d}$ and $\vec{c}$ is $\frac{\pi}{4}$, then $|10 - 3\vec{b} \cdot \vec{c}| + |\vec{d} \times \vec{c}|^2$ is equal to _____.

माना कि $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = 2\hat{i} + 2\hat{j} + \hat{k}$ और $\vec{d} = \vec{a} \times \vec{b}$ है। यदि $\vec{c}$ एक सदिश इस प्रकार है कि $\vec{a} \cdot \vec{c} = |\vec{c}|$, $|\vec{c} - 2\vec{a}|^2 = 8$ और $\vec{d}$ और $\vec{c}$ के बीच का कोण $\frac{\pi}{4}$ है, तो $|10 - 3\vec{b} \cdot \vec{c}| + |\vec{d} \times \vec{c}|^2$ बराबर है _____.

Ans. 6

Sol. $\vec{a} = \hat{i} + \hat{j} + \hat{k}$
 $\vec{b} = 2\hat{i} + 2\hat{j} + \hat{k}$
 $\vec{d} = \vec{a} \times \vec{b}$
 $= -\hat{i} + \hat{j}$
 $|\vec{c} - 2\vec{a}|^2 = 8$
 $|\vec{c}|^2 + 4|\vec{a}|^2 - 4(\vec{a} \cdot \vec{c}) = 8$
 $c^2 + 12 - 4c = 8$
 $c^2 - 4c + 4 = 0$
 $|c| = 2$
 $\vec{d} = \vec{a} \times \vec{b}$
 $\vec{d} \times \vec{c} = (\vec{a} \times \vec{b}) \times \vec{c}$
 Now,
 $\left(|\vec{a}| |\vec{c}| \sin \frac{\pi}{4} \right)^2 = ((\vec{a} \cdot \vec{c}) \cdot \vec{b} - (\vec{b} \cdot \vec{c}) \cdot \vec{a})^2$
 $4 = 4b^2 + (\vec{b} \cdot \vec{c})2(a^2) - 2(\vec{b} \cdot \vec{c})(\vec{a} \cdot \vec{b})$
 $4 = 36 + 3x^2 - 20x$
 Let $\vec{b} \cdot \vec{c} = x$
 $3x^2 - 20x + 32 = 0$
 $3x^2 - 12x - 8x + 32 = 0$

$$x = \frac{8}{3}, 4; \quad b.c = \frac{8}{3}, 4; \quad b.c = \frac{8}{3}$$

$$\text{Now, } |10 - 3b.c| + |d \times c|^2; |10 - 8| + (2)^2$$

$$\Rightarrow 6$$

60. Let α, β ($\alpha \neq \beta$) be the values of m , for which the equations $x + y + z = 1$; $x + 2y + 4z = m$ and $x + 4y + 10z = m^2$ have infinitely many solutions. Then the value of $\sum_{n=1}^{10} (n^\alpha + n^\beta)$ is equal to :

माना कि α, β ($\alpha \neq \beta$), m के वे मान है, जिनके लिए समीकरणों $x + y + z = 1$; $x + 2y + 4z = m$ और

$x + 4y + 10z = m^2$ के अनन्त हल हैं। तब $\sum_{n=1}^{10} (n^\alpha + n^\beta)$ का मान बराबर है :

- (1) 440 (2) 3080 (3) 3410 (4) 560

Ans. A

Sol.
$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 4 & 10 \end{vmatrix} = 1(20 - 16) - 1(10 - 4) + 1(4 - 2)$$

$$= 4 - 6 + 2 = 0$$

For infinite solutions

$$\Delta_x = \Delta_y = \Delta_z = 0$$

$$m^2 - 3m + 2 = 0$$

$$m = 1, 2$$

$$\alpha = 1, \beta = 2$$

$$\therefore \sum_{n=1}^{10} (n^\alpha + n^\beta) = \sum_{n=1}^{10} n^1 + \sum_{n=1}^{10} n^2$$

$$= \frac{10(11)}{2} + \frac{10(11)(21)}{6}$$

$$= 55 + 385$$

$$= 440$$